

Menu-Driven String Operations Using Functions

Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

Aim

To develop a menu-driven C program that performs uppercase conversion, string reversal, and palindrome checking using separate user-defined functions.

Algorithm

1. Start the program and display a menu with four choices.
2. Read the user's choice.
3. For choice 1, input a lowercase string and convert each character from a-z to A-Z by subtracting 32 from its ASCII value.
4. For choice 2, input a string, determine its length manually, and print the characters from the last position to the first.
5. For choice 3, input a string, find its length, compare characters from both ends, and decide whether it is a palindrome.
6. For choice 4, terminate the program.
7. If the choice is invalid, display an error message.
8. Repeat the menu until Exit is selected.

C Program

```
#include <stdio.h>

void toUppercase(void);
void reverseString(void);
void checkPalindrome(void);

int main(void)
{
    int choice;
```

```

do
{
    printf("\n--- STRING MENU ---\n");
    printf("1. Convert lowercase string to uppercase\n");
    printf("2. Reverse a string\n");
    printf("3. Check palindrome\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    getchar();

    switch (choice)
    {
        case 1:
            toUppercase();
            break;
        case 2:
            reverseString();
            break;
        case 3:
            checkPalindrome();
            break;
        case 4:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice. Try again.\n");
    }
} while (choice != 4);

return 0;
}

void toUppercase(void)
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }

    printf("Uppercase string: %s", str);
}

```

```

void reverseString(void)
{
    char str[100];
    int i, length = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    while (str[length] != '\0' && str[length] != '\n')
        length++;

    printf("Reversed string: ");
    for (i = length - 1; i >= 0; i--)
        printf("%c", str[i]);
    printf("\n");
}

void checkPalindrome(void)
{
    char str[100];
    int i, length = 0, flag = 1;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    while (str[length] != '\0' && str[length] != '\n')
        length++;

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - 1 - i])
        {
            flag = 0;
            break;
        }
    }

    if (flag == 1)
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");
}

```

Expected Output / Test Cases

Choice	Input	Expected Output
1	hello world	HELLO WORLD
2	computer	retupmoc

3	madam	The string is a palindrome.
3	hello	The string is not a palindrome.
4	-	Exiting program...

Result

The menu-driven C program was implemented using separate functions to convert lowercase text to uppercase without built-in conversion functions, reverse a string, check whether a string is a palindrome, and exit the program.